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A review of methods used to estimate initial landslide failure surface depths and volumes

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Abstract

This paper presents an overview of existing methods developed to characterize initial landslide failure surfaces and compute volumes based on surface information, when it is buried. This paper reviews the simplest models, such as volume estimation based on ellipsoids, and the information that can be obtained from surface displacement and morphology. A simple method to define the volume of a rockslide using the identified discontinuities is also outlined. The sloping local base level (SLBL), which is able to automatically obtain a possible failure surface on the basis of a digital elevation model (DEM), is described. The balanced cross section methods applied to landslides are illustrated and can give the depth of the failure surface. Using predefined assumptions, such as vector displacement parallel to the failure surface, we can obtain an approximation of the failure surface either along translational or circular slides, and this method can be improved by adding a mass balance equation. Recent work has also shown that using an inverse model, a mass balance equation and assumptions about the rheology enables estimation of the failure surface.

These methods are illustrated with examples, and they are applied to case studies and compared when the data available makes it possible. The advantages and disadvantages of the methods are discussed. This paper shows that the use high-resolution topography and interpretation of the morphology is underexploited to extract information about failure surfaces geometry.

Key words: landslide, failure surface, volume, surface area, depth.

1 Introduction

The characterization of failure surface and volume of landslides is important to understand their behaviors and their potential effects on objects at risk such as buildings, roads, infrastructures. In fact, volume is one of the most important parameters controlling the distance of propagation, the affected area, and the potential damages (Corominas, 1996; Jakob, 2005; Jakob et al., 2012; von Ruette et al., 2016). The product of the surface area and the average depth of landslides is often used to estimate their initial volumes, or it is used to give a range of volume calculated from a range of depth (Guzzetti et al., 2009). But this can be refined by different techniques based on surface information. The aim of this paper is to present an overview of methods, which can be used to estimate the depth and/or

the shape of the initial failure surface or the initial volume of landslides, when the failure surfaces are not visible, but the limits of the landslide are identifiable. This means that this paper does not tackle the deposits shapes and volumes. Most methods rely on expert knowledge and synthesize the available data of surface morphology, boreholes, geophysics interpretation, etc. In some cases, when structures (faults, bedding, etc.) are present, the construction is very close to the classic geological cross section (Chigira et al., 2013). Only a few methods are based on a systematic approach.

Estimating failure surfaces depths or shapes along profiles or in 3D or volumes —without subsurface information— is often based on sparse data (Hutchinson, 1983). Only a few methods have been proposed to estimate surface depths and/or volumes, such as half-ellipsoid approximation (WP/ WLI, 1990; Cruden and Varnes, 1996) or simple area-thickness relationships (Guzzetti et al., 2009, Larsen et al, 2010). Other methods have been proposed to infer cross sections using morphology or surface displacements (Hutchinson, 1983; Carter and Bentley, 1985) based on classic methods, such as electronic distance measurement (EDM), inclinometers, crackmeters, and extensometers (Hutchinson, 1983).

However, nowadays new data types and increasing computer power provide new avenues of method development. High-resolution digital elevation models (HR-DEMs) or point clouds (Girardeau-Montaut, 2006) provide very detailed information about the surface morphology and are useful for interpreting the subsurface geometry. This information can be obtained from laser scanners (Jaboyedoff et al., 2012) or structure from motion (SfM) photogrammetric models (Eltner et al., 2017). Sequential acquisitions also provide a way to obtain surface displacement fields (Travelletti et al., 2012; Aryal et al., 2015). Such movement data can also be obtained by methods such as interferometric synthetic aperture radar (InSAR) (Rott and Nagler, 2006; Strozzi et al., 2015; Nikolaeva et al., 2014; Wasowski and Bovenga, 2014) and image cross correlation, in addition to Global Navigation Satellite System (GNSS) (Travelletti et al., 2012). Many methods are currently available, but they are still not widely utilized.

Our overview of the existing methods starts with the simplest shapes, such as ellipsoids, or the method of transversal cross-sectional area for channelized landslides. Then, we present the classic relationships between surface area and volume. The use of splines in different ways is presented to determine transversal cross sections. The sloping local base level (SLBL) is also presented, which allows us to deduce a possible surface of failure based on a DEM, a second derivative value of this surface and the landslide contour. When the movement at the scarp is measurable, it is possible to deduce the thickness using either circular failure, graben structures or balanced cross sections. Using the surface velocity vectors of displacement, assuming they follow the same trend as the failure surface, the depth of the failure surface can be deduced along profiles. Adding conservation of mass to such an approach permits improvement of the method.

Some attempts to infer failure surfaces have recently been made to take advantage of new data types (Booth et al., 2013; Nikolaeva et al., 2014; Aryal et al., 2015). These exploratory works have shown that the potential for development exists, which is why an overview of the existing methods and some ongoing developments are presented here.

Some examples are given, and comparisons of the methods are shown to evaluate their validity. Some attempts to quantify the uncertainty are proposed, including the probability distribution of the failure surface depth. Most methods reviewed here are based on morphometric characterizations,

displacement observations, and interpolation techniques normally without subsurface information, leading to the estimation of failure surfaces shapes in 3D or along cross-section, or depth under various assumptions. This paper extends the work on this topic, which was previously published as a conference paper (Jaboyedoff et al., 2015). Because many different formulas are presented, a list of variables and symbols are given in Table 1.

Table 1: List of variables and symbols used.

Variables	Description
a (a_L , a_w)	Second derivative of a parabola and coefficient of the square (along the length and width of the landslide)
A , B , C	Coefficients of spline polynomials (when together) or triangle corners
A_0	Initial horizontal surface area of a vertical prism
A_1 ,	Surface area in cross section removed by the movement at the scar or horizontal surface area of a vertical prism
A_2	Rectangular surface area in cross section indicating the material crossing a surface perpendicular to the failure surface
A_h	Horizontal surface area of a landslide
A_s	Surface along the slope
C	Tolerance for the SLBL
ce	Expansion coefficient
D	Depth of the failure surface perpendicular to the slope
D_d	Maximum depth of the deposit perpendicular to the slope and half axis of the ellipsoid or depth of the elliptic paraboloid
D_r	Maximum depth of the surface of rupture perpendicular to the slope and half axis of the ellipsoid or depth of the elliptic paraboloid
e	Ratio of z_{max} and L_{rh}
f	Coefficient linking surface velocity and average and depth average velocity
$f()$	Function or probability distribution function or function averaging elevation for SLBL
$F()$	Repartition function
g	Freefall acceleration
h	Vertical thickness
H_0	Height of a vertical prism
k	Constant coefficient of the power law for the surface area-volume relationship
l	Assumed movement along the sliding surface
L_d	Length of the deposit along the slope and axis of the ellipsoid or length of the elliptic paraboloid
L_i	Length of reach i for debris flows and channelized landslides
L_r (L_{rh})	Length of the surface of rupture along the slope and axis of the ellipsoid or length of the elliptic paraboloid (projected horizontally)
M	Matrix that contains the coefficient of the horizontal surface velocity
m	Landslide surface elongation
n	Exponent of the Herschel-Buckley law
r_0 , $r(\theta)$	Radius of spiral slip surface at the top of the slope, at the angle θ
R	Radius of a circular failure
S_i	Section area i of the available sediments
t	Time or iteration number
t_v , t_{max}	Mean vertical thickness, maximum vertical thickness

$u_s, \bar{u}, u_{sh}$	Surface velocity, depth averaged velocity, horizontal surface velocity
u_x, u_y, u_z	Surface displacement velocity vector components
V	Volume
V_{ep}	Volume of the elliptic-paraboloid defined by the initial failure surface
V_{de}	Volume of the “displaced mass” or of the “deposit” using half-ellipsoid
V_{init}	Initial volume of the debris flows or channelized landslides
V_{lat}	Lateral volume input for the debris flows or channelized landslides
V_{re}	Volume of the half-ellipsoid defined by the initial failure surface
w	Width of the landslide failure surface and axis of ellipsoid or width of the elliptic paraboloid
w_i	The width of the scouring surface for debris flows or channelized landslides
W_d	Width of the deposit and axis of ellipsoid
x, x_i	Value along any chosen x axis
x', z'	Coordinates along and perpendicular to the slope
z, z_0, z_{mp}	Value of the vertical axis downward, vertical position of surface topography, most likely vertical depth of the failures surface
z_{i0}	Depth of the expected scouring surface at the center of the bed from a straight line joining the side of a torrent prone to debris flow
z_{i1}	Depth at the center of the bed from its side of a torrent prone to debris flow
$z()_{ij}, z_{ij}$	Vertical coordinates on a raster
z_{max}	Maximum vertical thickness of landslide calculated from a straight line joining the upper scarp and the toe of the initial failure surface or deeper expected failure surface vertical depth
$z_n(), z_{temp}()_{ij}$	Average altitudes of a set of neighbouring points of a grid, temporary value of the altitude during an iterative calculation
Δh	Length of the scarp
Δx	Grid cell size
α	Exponent of the power law for the surface area-volume relationship
β	Average slope angle
δ, δ'	Angle of rotation or a rotational landslide or change in angle of a failure surface, angle of aperture of a circular landslide
ϕ	Friction angle
θ	The angles from the top of the slope for spiral slip surface
μ_n	Dynamic viscosity
ρ, ρ_0, ρ_l	Density and initial and later landslide material density
τ	Shear stress
ξ	Standardized variable for the spline calculation
ψ	Slope angle of a vertical wedge side (graben)

2 Assessing slope failure surface shape and/or volumes based on cross-sections

2.1 Classic methods

These methods consist of producing cross sections based on surface data and expert knowledge. When using geological constraints, the methods of construction are very similar to geological cross section methods based on structural projections. Using a series of parallel cross sections can provide a 3D surface and an accurate volume estimation by summing the multiplication of the surface area of

the cross sections by the sum of the half-distances to their two neighbors' profiles (Chigira et al., 2013).

2.2 Standard failure surface profile shapes

Circles are used to describe profiles of failure surfaces. These circles are mostly linked to the classic slice methods (Duncan and Wright, 2005; Terzaghi et al. 1996), which can include a search for the lower factor of safety for a given slope (Rocksciences, 2018). Skempton and Hutchinson (1969) indicate that, for rotational slides, the ratio D_r/L_r (maximum depth perpendicular to the slope/length of the landslide) ranges between 0.15 and 0.33, which corresponds to circular arcs ranging δ' from 65° to 135° . In addition, as noted by Varnes (1978), smaller values of the D_r/L_r ratio are linked to shallow landslides. The sections obtained by these methods can support the estimation of volumes (Dewitte and Demoulin, 2005), using the hypothesis about the third dimension (see sections 3.2, 4.1, 4.2).

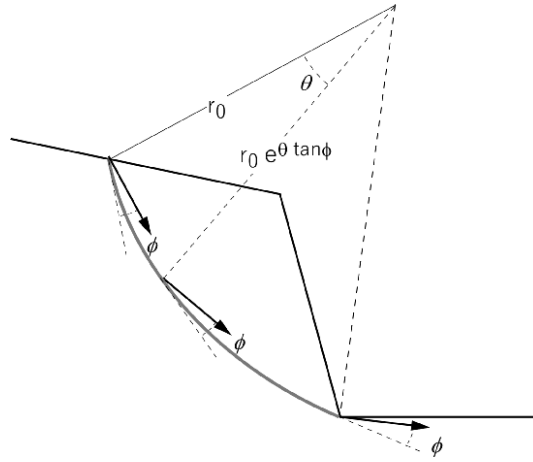


Fig. 1: illustration of the log spiral failure profile including the relationship with the friction angle (modified after Chen, 1975)

Another type of profile is the log spiral, which is considered by many authors to be the profile of the failure surface in homogeneous material (Terzaghi et al., 1996; Duncan and Wright, 2005; Chen, 1975):

$$r(\theta) = r_0 e^{\theta \tan \phi} \quad (1)$$

where $r(\theta)$ is the radius of the spiral slip surface at the angle θ from the top of the slope where the radius is equal to r_0 . ϕ is the friction angle along the slip surface (Fig. 1). The characteristics of the log spiral show that the only resisting force comes from the cohesion. The log spiral minimizes the failure energy (Chen, 1975) along the failure surface and has recently been used to model the failure surface of the Oso landslide (Iverson et al., 2015). Furthermore, the log spiral has been used to establish a relationship between landslide volume and surface area that fits the observations for landslides (Klar et al., 2011; see section 3.3) and for retrogressive cliff evolution promoted by landsliding (Utili and Crosta, 2011).

3 Assessing volumes based on landslide cross-sections, widths, lengths or surface areas

3.1 Half ellipsoid or elliptic paraboloid failure surface shapes

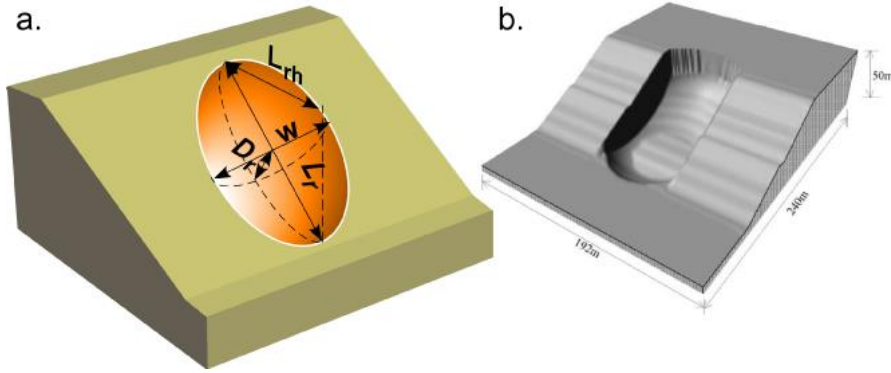


Fig. 2: Synthetic way to define volumes of landslides by the failure surface. A. D_r is the depth of the half ellipsoid (landslide), and w and L_r are the width and length, respectively, in the slope direction. L_{rh} is the length projected horizontally (modified after Cruden and Varnes, 1996). B. Use of half ellipsoid for create a surface of failure (see text) (from Xie et al., 2004).

The simplest method used to assess landslide volume ($V = A_h \times t_v$) assume a mean vertical thickness (t_v) multiplied by the landslide horizontal surface area (A_h), this is a suitable approach for the shallow landslide affecting soil over hard rock.

The method classically proposed is to assume a semiellipsoid shape (which assumes a planar topographic surface) for the failure or rupture surface using a coordinate system defined along the slope oriented to the major axis of the landslide (WP/ WLI, 1990; Cruden and Varnes, 1996):

$$V_{re} = \frac{1}{6} \pi \times D_r \times w \times L_r = \frac{2}{3} A_s \times D_r = \frac{2}{3} A_s \times D_r = \frac{2}{3} \frac{A_h}{\cos \beta} \times D_r \approx \frac{2}{3} A_h \times z_{\max} = \frac{1}{6} \pi \times w \times L_{rh} \times z_{\max} \quad (2)$$

where w , L_r , and A_s are the width (assumed close to horizontal), length, and surface area of the rupture along the slope, respectively, D_r is the depth perpendicular to the slope, and β is the average slope angle (Fig. 2a). The surface area of the ellipsoid $\pi \times w \times L_r / 4$ can be replaced by A_s or the projection of A_h . In addition, assuming a vertical depth $z_{\max} \approx D_r / \cos(\beta)$ the volume can be calculated using A_h . Dewitte and Demoulin (2005) combined the ellipsoid methods with the value of the D_r/L_r ratio by Skempton and Hutchinson (1969) to obtain volume estimation ranges. Because the semiellipsoid possesses nearly vertical limits, it can be replaced by elliptic paraboloids, which gives:

$$V_{ep} = \frac{1}{8} \pi \times w \times L_r \times D_r = \frac{1}{2} A_s \times D_r = \frac{1}{2} \frac{A_h}{\cos \beta} \times D_r \approx \frac{1}{2} A_h \times z_{\max} = \frac{1}{8} \pi \times w \times L_r \times z_{\max} \quad (3)$$

See additional details on this formula in section 5.1.2. V_{re} and V_{ep} corresponds to the volume defined by the initial failure surface, with no change in the volume of the material. Cruden and Varnes (1996)

pointed out that the volume V_{de} of the “displaced mass” or of the “deposit” is related to V_{re} by the relationship:

$$V_{de} = \frac{1}{6} \pi D_d \times W_d \times L_d = ce \times V_{re} = \frac{4}{3} V_{re} = \frac{2}{9} \pi \times D_r \times w \times L_r \quad (4)$$

The deposit volume is also assumed to be ellipsoidal. This remark emphasizes that volume estimation must be clearly defined and depending on whether we consider the failure surface or the displaced volume, the expansion factor ce value proposed by these authors is 1.33. This remark underlines that the estimation of the deposit volume is a subject in itself, which includes the expansion factor that will be not be tackled further in this paper.

These approaches have been generalized using GIS tools and additional criteria. Marchesini et al. (2009) fit ellipses on the contours of landslides. The major axis is chosen along the slope with a length equal to the length of the contour in that direction. The center is chosen as the centroid of the landslide surface area. The second axis is taken perpendicular to the slope with a length equivalent to the width of the landslide contour. The third axis, the thickness, is perpendicular to the slope and calculated by minimizing the safety factor by varying the length of this third axis. The safety factor is calculated using a simplified probabilistic Janbu analysis applied in 3D using the surfaces defined by the ellipsoids. The parameters such as friction angle, cohesion, and water table level are predefined. This method is based on the work by Xie et al. (2004), which used additional constraints such as geological subsurface layers (which can cut the ellipsoids) but no predefined contours of the landslides, which were chosen randomly (Fig. 2b). Nikolaeva et al. (2014) used dislocation theory to find the depth, length, position, dip, dip direction and slip component of a rectangular sliding plane. Genetic algorithm was used create a model that fits the InSAR data. Then this sliding plane is linked to the surface by fitting an ellipsoid to define the volume.

3.2 Transversal cross section areas-volume for debris flows or channelized landslides

The volume estimation of potential debris-flows or channelized landslides is mainly performed through an estimate of the available volume of sediment in the channels which is available by scouring, and some lateral contributions can be considered, i.e. material that can come from the other landslides affecting the channel banks (Hungr et al., 2005).

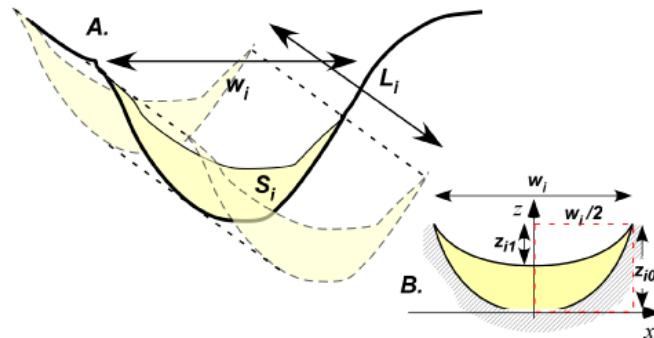


Fig. 3: Synthetic way to define volumes of material in a torrent channel (A) using observed cross sections and (B) based on parabolic profiles.

The classic method to assess these potential volumes is to perform surveys, estimate the transversal cross-sectional area S_i of the available sediments, then multiply the S_i by the length L_i of the reach centered on the cross-section (Fig. 3) and add the initial volume V_{init} and the lateral input V_{lat} (Hungre et al., 2005):

$$V_{total} = V_{init} + \sum_{i=1}^n S_i L_i + \sum_{j=1}^m V_{lat}^j \quad (5)$$

The transversal section can be replaced by polynomial functions to estimate the maximum scouring potential when only a few data are available. Using a parabola as the bed shape section and for the erodible surface, we show by integration of the parabolic profiles that section S_i can be replaced by:

$$S_i = \frac{2}{3} w_i \times (z_{i0} - z_{i1}) \quad (6)$$

where z_{i1} is the depth at the center of the bed from a line joining its both sides and z_{i0} is the depth and w_i the width of the expected scouring surface. Currently, the estimation of debris flow deposits is mainly performed by HR-DEM differences (Jaboyedoff et al., 2012; Arattano et al., 2015).

3.3 Surface area-volume relationship

Many authors have attempted to estimate the volume based on the horizontal surface area (A_h) or along the slope (A_s) of landslides using a power law function (Fig. 4):

$$V = k \times A_s^\alpha \quad (7)$$

where k is a constant and α is an exponent. These relationships are often based on assumptions (Table 2; Fig. 4). Hovius et al. (1997) deduced their relationship using a width to length aspect ratio of 2 (based on observation) and the landslide thickness and linked it to the horizontal surface area by $t_v = 0.5 \times (A_h)^{0.5}$. Guzzetti et al. (2009) obtained their relationship using a worldwide catalog of landslides inventories that was compiled from the literature. The estimates of volumes of both approaches are expert based, because most of the time the thickness is estimated by the geologist without subsurface information. Such power laws are supported by an analysis based on limit equilibrium analysis using log spiral or circular profiles, showing that α theoretically ranges from 1.32 to 1.39 (Kalderon-Asael et al., 2008; Klar et al., 2011), which agrees with the values deduced from inventories. These values are similar to the average of all values of table 2 $\alpha=1.372$ ($k = 1.267$) and of the world inventory from Guzzetti et al. (2009), by removing the values of shallow and submarine landslides and from those of Korup (2006) from our average calculation, we obtain an equivalent value $\alpha=1.362$ ($k = 0.288$). The α values for shallow landslides are clearly lower than that for other types of landslides, with an average value of 1.14 equivalent to the results of Larsen et al. (2010). The studied submarine landslides seem to also have a lower α . This finding indicates that the D_r/A_s ratio of shallow and submarine landslides decreases rapidly with increasing surface area. To summarize, the values provided by Guzzetti et al. (2009), i.e., $\alpha = 1.450$ and $k = 0.074$, appear to be excellent, but reasonable ranges for large landslides are $\alpha = 1.25$ to 1.50 and $k = 0.05$ to 0.77 . However, there is a clear trend (an exponential relationship) of increasing α when k decreases (Fig. 4b). It is also interesting to notice

that for range of volumes above 10^5 m^3 , the results of Guzzetti et al. (2009) and from Larsen et al. (2010) are very similar considering the large variability, even though α differs by 0.1 and k by 0.408, indicating that the exponent is to interpret with caution.

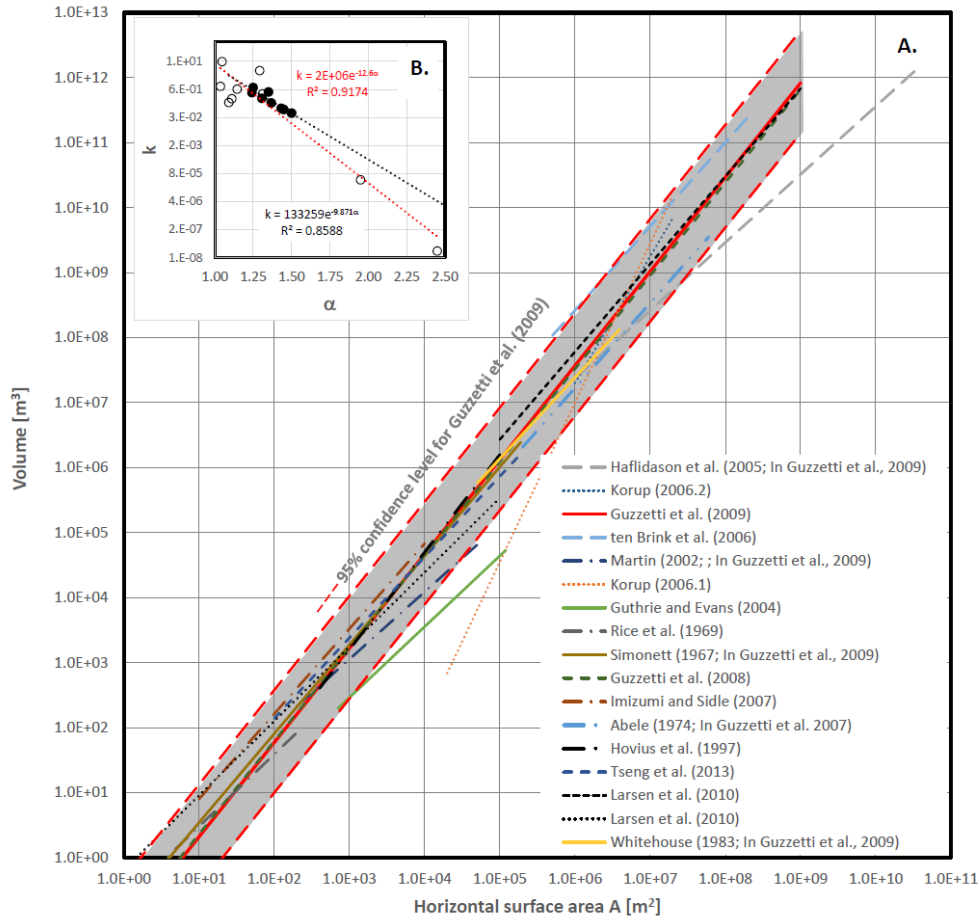


Fig. 4: (A) Surface area-volume power law from different authors. The continuous red line is the synthesis and the dashed red lines plus the gray shaded area indicates the corresponding 95% confidence level of Guzzetti et al. (2009). (B) Exponential relationships between α and k based on all values (in red) and excluding the values for shallow and submarine landslides ($\bullet$ = selected points for the black regression and o = points additional used for the red regression).

Table 2: Surface area-volume power law from constants by different authors (modified from Guzzetti et al., 2009).

Sources	Type	β	k	Number of Landslides	Lower value A_s [m ²]	Higher value A_s [m ²]
<i>Abele (1974; In Guzzetti et al. 2009)</i>	Deep-seated landslide in European Alps (>.2 km ²)	1.307	0.242	63	2.0.E+05	6.0.E+07
<i>Guthrie and Evans (2004)</i>	Shallow landslide Vancouver Island, BC, Canada	1.091	0.155	124	7.0.E+02	1.2.E+05
<i>Guzzetti et al. (2008)</i>	Upper Tiber River basin, central Italy	1.432	0.084	539	2.0.E+00	1.0.E+09
<i>Guzzetti et al. (2009)</i>	World synthesis	1.450	0.074	677	2.0.E+00	1.0.E+09
<i>Haflidason et al. (2005; In Guzzetti et al., 2009)</i>	Submarine landslides in coastal Norway	1.047	12.27 3	65	3.0.E+05	3.9.E+10
<i>Hovius et al. (1997)</i>	Southern Alps New Zealand	1.500	0.050	4984	4.0.E+02	1.0.E+05
<i>Imaizumi and Sidle (2007)</i>	Landslide in small Japanese catchment	1.310	0.390	51	1.0.E+01	1.0.E+04
<i>Korup (2006.1)</i>	Landslides SW New Zealand (>.02 km ²)	2.450	1.995E- 08	34	2.0.E+04	2.0.E+07
<i>Korup (2006.2)</i>	Landslides SW New Zealand (>1 km ²)	1.950	3.991E- 05	23	1.0.E+06	2.0.E+07
<i>Martin (2002; ; In Guzzetti et al., 2009)</i>	Shallow landslide BC-Canada	1.036	0.880	615	2.0.E+02	5.2.E+04
<i>Rice et al. (1969)</i>	Shallow landslide Ca, USA	1.110	0.234	29	2.1.E+00	2.0.E+02
<i>Simonett (1967; In Guzzetti et al., 2009)</i>	Landslide in New Guinea	1.368	0.148	207	2.3.E+00	1.9.E+05
<i>ten Brink et al. (2006)</i>	Submarine	1.292	4.655	201	5.0.E+05	2.0.E+08
<i>Whitehouse (1983; In Guzzetti et al., 2009)</i>	Deep-seated landslides Central Southern Alps	1.250	0.769	45	5.0.E+04	3.9.E+06
<i>Larsen et al. (2010)</i>	Bedrock landslides	1.350	0.482	604	1.0.E+05	1.0.E+09
<i>Larsen et al. (2010)</i>	Soil landslides	1.145	0.644	1614	1.0.E+00	1.0.E+05
<i>Tseng et al. (2013)</i>	Typhoon Marakot landslide (DTM difference)	1.242	0.452	371	1.0.E+02	2.0.E+05

4 Transversal cross section based on morphology and splines

4.1 Empirical transversal cross section

This method is an empirical method that has been developed for teaching purposes by one of the authors. The method allows easy creation of a transversal cross section of a landslide by hand when the profile is completely unknown. In fact, when using a drawing software, this method corresponds to the use of splines, adjusting the positions using the endpoints which implicitly varies first and second derivatives. The process is as follows (Fig. 5):

1. Draw lines tangent to both edges of the failure surface (points P_1 and P_2), which can be considered the lower potential sliding surface.
2. Form a triangle P_1 - P_2 - P_0 , where P_0 is the intersection of the tangents of the slope at points P_1 and P_2 .
3. Draw the 3 bisectors of this triangle.
4. Manually construct a possible curved line joining the two outer points P_1 and P_2 that is tangent to lower triangles sides at both end in P_1 and P_2 and is parallel to the upper triangle side at the intersection point of the bisectors (P_3). This represents the expected failure surface.
5. Repeat the point 4 procedure for two additional lines: Construct a line tangent to the bisectors from point P_1 to P_2 , going through the point P_5 , locate on the “vertical” bisector in the at the middle distance $d'/2$ from the bisector intersection and the upper side of the triangle. In P_5 the tangent is parallel to the upper side of the triangle. Construct a line below the “failure surface” from the points P_1 and P_2 tangent to a line making an angle of the half of internal angles at P_1 and P_2 and joining P_4 located at the half of the distance $d/2$ between the lower corner P_0 and the intersection of the bisectors. The tangent P_4 must be parallel to the upper side of the triangle.

The zone between these two new lines (in gray in Fig. 5) is the possible location of the slip surface.

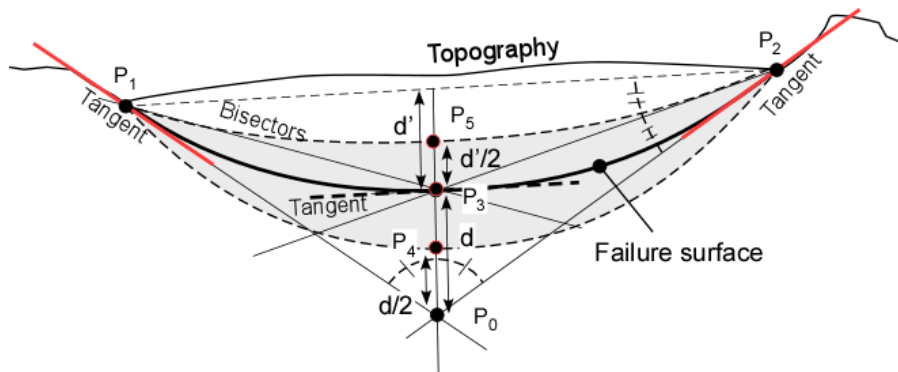


Fig. 5: Illustration of the empirical method of bisectors. The gray area indicates the most likely location of the sliding surface. It can be assumed that the surface passes along a path that may be more complex.

4.2 Transversal section using spline calculation

Splines are lines or surfaces that are commonly used in 3D geological models (de Kemp, 1999). Splines can be used to obtain reliable estimates of failure surface profiles based on simple splines in one dimension using geomorphic features (Jaboyedoff and Derron, 2015). For example, a topographic section (x - z) where the surface of failure starts at point x_1 and the end at x_2 is

identified, and the derivatives or slope angles are known in the points within the cross-section plane. Thus, a third-order polynomial (1D-spline) can be drawn using (Bartels et al., 1987):

$$z(\xi) = A + B\xi + C\xi^2 + D\xi^3 \text{ with } \xi = \frac{x - x_1}{x_2 - x_1} \quad (8)$$

To illustrate this method, we used the La Frasse landslide as an example (Noverraz and Bonnard, 1988). The results are compared to the interpretation by the geologist (DUTI, 1986), which was based on boreholes, and they show very good agreement (Fig. 6).

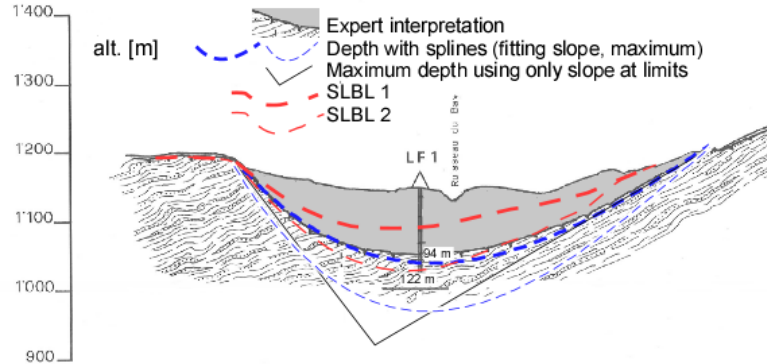


Fig. 6: Comparison of the spline method and SLBL method with the interpretation by the geologist (modified after DUTI, 1986 and Jaboyedoff and Derron, 2015).

5 3D failure surfaces based on morphology

5.1 Sloping local base level (SLBL)

5.1.1 Principle of SLBL

SLBL is a method that provides a 3D surface from a DEM based on the assumption that a failure surface can be extrapolated from the limits of the unstable mass assuming some geometrical properties about its shape, such as border, slope, and second derivative. The SLBL method can be easily parameterized and fairly automatized.

The principle of SLBL is rather simple. First, the perimeter of the instability must be defined. Then, by an iterative process, the SLBL method “numerically” determines the volume within the defined limits from a gridded DEM ($z(t)_{ij}$) as follows (Jaboyedoff et al., 2004, 2009).

For each grid node of the DEM and at each iteration t , the SLBL method estimates an “average” $z_{temp}(t)_{ij}$ of all the altitudes ($f(z_n(t-1))$), where z_n means a set of neighbors of the previous iteration $z(t-1)_{ij}$ of the points minus a positive constant C (tolerance), i.e.,:

$$z_{temp}(t)_{ij} = (f(z_n(t-1)) - C) \quad (9)$$

This process is performed either by a simple average of a given number of neighbors or by fitting a surface. This method creates a new grid considering that if $z_{temp}(t)_{ij} < z(t-1)_{ij}$, then $z(t)_{ij} = z_{temp}(t)_{ij}$; otherwise, the value is unchanged. An additional condition can be added that the $z(t)_{ij}$ value is not below a limit defined by another DEM or assuming a limiting slope angle. The iteration is repeated

until all the differences over the grid ($z_{temp}(t-1)_{ij} - z(t)_{ij}$) are smaller than a given threshold. This method can also be based on the total volume change between two successive surface iterations.

Note that C depends on the grid cell size Δx , because it is linked to the numerical second derivative involving Δx . In addition, as C is assumed constant all over a landslide, it can be related to the ratio e of its maximal vertical depth z_{max} and length L_{rh} (or width) of the landslide. z_{max} is deduced from a theoretical cross section of the failure surface (for instance manually drawn) by joining a straight line from its two intersections with the topography and measuring its larger distance to this line. If the profile is assumed parabolic of equation $ax^2 + bx + c$, C is proportional to the second derivative $2a$, and assuming b and c are equal to zero, thus $a (L_{rh}/2)^2 = z_{max}$. Thus, it can be shown that (Jaboyedoff et al., 2019):

$$C = a \Delta x^2 = 4 \frac{z_{max}}{L_{rh}^2} \Delta x^2 = 4 \frac{e}{L_{rh}} \Delta x^2 \quad (10)$$

L_{rh}^2 can be replaced by the squared geometrical mean of w and L_{rh} i.e. $C = 4 z_{max} \Delta x^2 / A_h$, assuming $A_h \approx w \times L_{rh}$. Note that it is also possible to fill holes or valleys by changing the sign of C and reversing the condition in point 2.

5.1.2 Equivalence between the SLBL and elliptic paraboloid

As the principle of the SLBL is to use a constant second derivative, its application to an elliptical landslide will provide an elliptic paraboloid, as we verified. Thus, we can develop a very simple model to link the volume to the C value and using the elongation $m = L_{rh}/w_r$ of the landslide deduced from the horizontal projection of its contour which is assumed the ratio of the ellipsoid axis of the landslide (Fig. 2). Therefore, the volume does not depend on the slope angle. For an elliptical landslide, the a value depends on the direction of the chosen axis. The SLBL is calculated based on the average of a , but it depends on the axis; thus, $a = (a_L + a_w)/2$, where a_L is the second derivative along the long axis and a_w is that along the short axis. If the origin of the coordinates is located at the center of the ellipse and the maximum vertical thickness is z_{max} :

$$z_{max} = a_L \frac{L_{rh}^2}{4} = a_w \frac{w^2}{4} = a_L m^2 \frac{w^2}{4}, i. e., \frac{a_L}{a_w} = \frac{1}{m^2} \quad (11)$$

Because a is the average:

$$a = \frac{a_L}{2} (m^2 + 1) \quad (12)$$

Using the above equations to provide relationships for C , z_{max} , and a_L , the volume of the elliptic paraboloid is given by:

$$V_{ep} = \frac{1}{8} \pi L_{rh} w z_{max} = \frac{\pi}{16} L_{rh}^4 \frac{a}{m(m^2+1)} = \frac{\pi}{16 \Delta x^2} \frac{C L_{rh}^4}{m(m^2+1)} \quad (13)$$

This equation shows that for a simple geometry, the V_{ep} is linearly related to C or a . This equation provides a way to estimate the parameter used to calculate the SLBL. In addition, if the aspect ratio m can be evaluated as well as the second derivative of the failure surface, which, for instance, can be defined by the shape of the scarp, this method can improve the estimation of the volume calculated by a simple half ellipsoid and better determine the value of z_{max} .

5.1.3 Example of the reconstruction of a failure surface

From 2 to 5 September 2011, the Typhoon Talas hit the Kii Peninsula (Japan), triggering 70 deep-seated landslides (Chigira et al., 2013). Comparing 1 m LiDAR DEMs that were acquired before (pre-DEM) and after (post-DEM) the event permitted the reliability of the SLBL method to be tested (Jaboyedoff et al., 2019).

Three different reconstructions of the failure surfaces are proposed using a 5 m DEM resolution (Jaboyedoff et al., 2019), for the Nagatono landslide which occurred in a formation made of sandstone blocks in a mudstone matrix. First, the limit of the potential landslide was defined based on the hillshade of the pre-DEM 1 m (Fig. 7). Based on a handmade cross section, the C value was deduced using the ratio e . The initial volume obtained is $3.94 \times 10^6 \text{ m}^3$, which is very similar to the result obtained by Chigira et al. (2013) ($4.1 \times 10^6 \text{ m}^3$) using cross-section method (section 2.1). It was easy to adjust C to obtain exactly the same volume. Using the contour of the landslide based on the difference of the two HR-DEM, we obtain an area that is slightly narrower than the previous solution. By increasing the C value, we easily obtain a very good fit of the failure surface in the upper part where it is free of deposits with a volume of $4.02 \times 10^6 \text{ m}^3$. The failure surface is not perfectly matched, most likely because of preexisting structures inducing weaknesses, but on average, the fit is good.

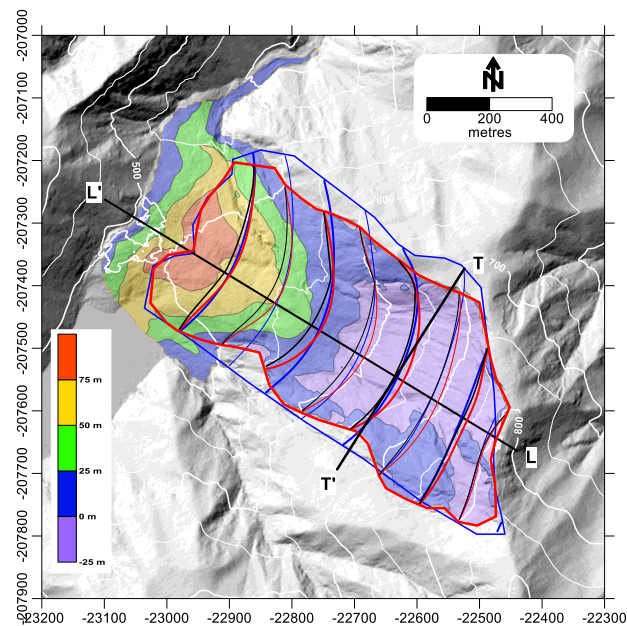


Fig. 7: Comparison of the SLBL results with a post-landslide DEM (from Jaboyedoff et al., 2019). Hillshade from the 1 m post-DEM of the Nagatono landslide (Kii Mountain, Japan), with contour lines displayed in red, and a failure surface altitude deduced using the contours of landslides obtained by comparing pre- and post-DEMs. A similar solution with a different C value is shown in black. The result shown in blue was obtained using the pre-DEM to create the potential limits based on only surface morphology. The colored area refers to the thickness deduced from the subtraction of the post-DEM from the red scenario.

5.2 Volumes of rock instabilities controlled by discontinuities

In hard rock, the major discontinuities shape most of the instability limits, and they are often outcropping. Consequently, these discontinuities can be used to extrapolate the failure surfaces to deduce the potential mechanism of failure and their volumes. Using a point cloud or a DEM, the extent of the instability can be identified based on outcropping discontinuities. Extrapolating the

subsurface discontinuities can be performed by either manually fitting planes on the discontinuities or by extrapolating discontinuity traces using structural analysis from either field surveys or point cloud data. The procedure based on a gridded DEM is as follows (Fig. 8):

1. Create the contour of the rock instability using the basal and rear-bounding surfaces that delimit the instability;
2. Extrapolate the discontinuity to delineate the volume; and
3. Compute the volume and thickness map, which is usually based on DEM differences within a GIS environment. Or using specialized software for point clouds treatment such as CloudCompare (2018; Girardeau-Montaut, 2006), which permits to calculate volume between surfaces or point clouds.

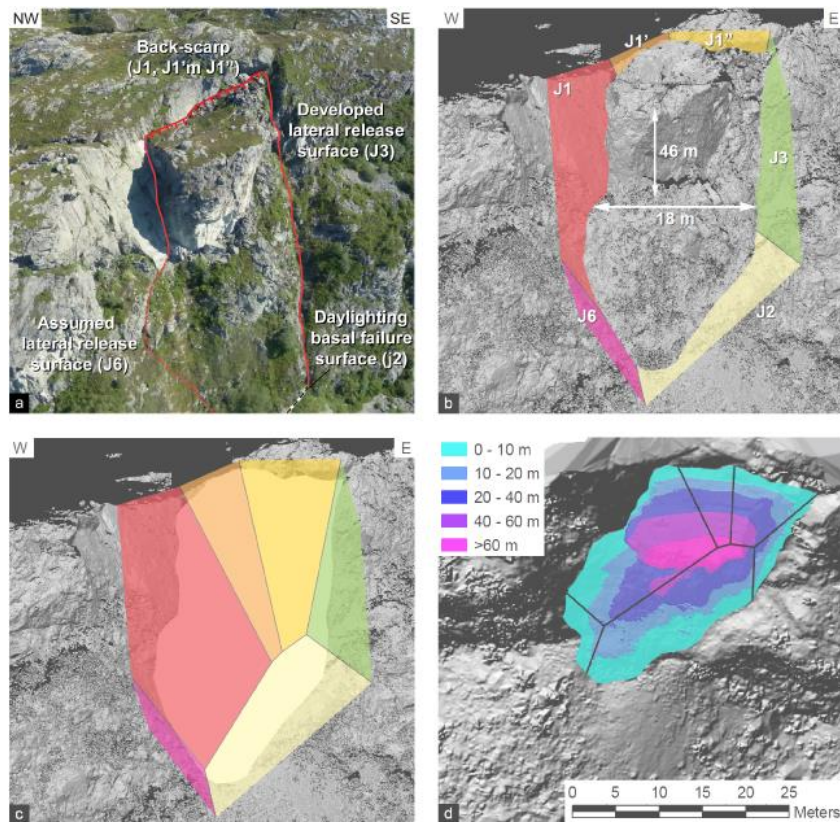


Fig. 8: Construction in 3D of the basal failure surface of the unstable rock slope Lausefjellet in western Norway: a) overview photograph of the instability (photograph: L. Fischer, Geological Survey of Norway); b) planar surfaces used to model the failure surface (3D view to the north); c) modeled failure surface under the instability; d) map view of elevation differences between the present topography and modeled failure surface (represented by black intersection lines between the planar surfaces in b) and c)) (modified from Oppikofer et al., 2016).

Such an approach often provides the maximum volume that can be involved in a rock failure because, in most cases, the failure surface is more complex due to the presence of rock bridges and because minor discontinuities are often also involved in shortcutting some angular shapes. However, in most cases, the minor discontinuities will be contained within the volume defined by the major structures.

6 Methods based on surface displacements or velocities in longitudinal cross section

6.1 Longitudinal cross section based on apparent displacements or velocities

Observations of the total surface displacements or the velocity vector field can lead to information on landslide depth and to the construction of longitudinal profiles of failure surfaces whether or not they are circular. As an example, the total displacement in the scarp of a translational slide can provide the thickness of the landslide.

6.1.1 Scarp length and rotation

When the head scarp of a rotational slide is visible and the rotation at the surface is measurable, the radius (R) can be estimated (Fig. 9) by measuring the length of the scarp Δh and the angle of rotation δ (Jaboyedoff et al., 2009):

$$R = \frac{\Delta h}{2 \cdot \sin \frac{\delta}{2}} \quad (14)$$

The location of the center of the circle can be located either by fitting the circle on the scarp or using the scarp and an estimate of the bottom of the initial failure surface, which is usually located below the deposit.

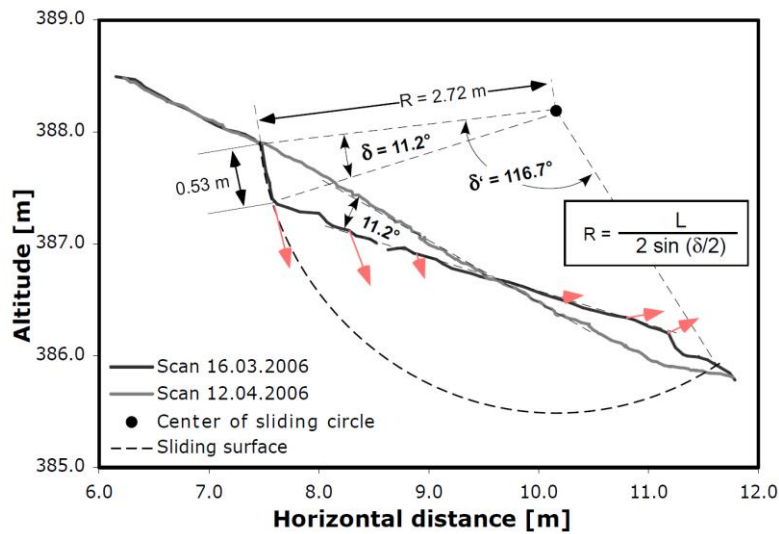


Fig. 9: Example calculation of an observed landslide using terrestrial laser scanners. Note that the result of fitting a sphere on the head scarp surface is very close to the result of eq. 14 with $R = 2.72$ m (modified after Jaboyedoff et al., 2009).

6.1.2 Using the trace of a wedge

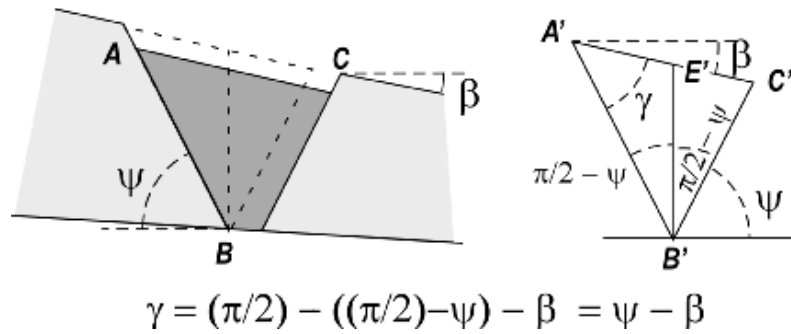


Fig. 10: Illustration of the geometry used to solve the problem of the double wedge (modified after Cornforth, 2005).

Cornforth (2005) proposed a way to estimate the maximum thickness of a translational slide by including a double wedge (“graben”) in its head scarp. The maximum vertical thickness t_{max} can be estimated assuming that the width of the wedge is close to the undeformed state and that the wedge scarp and reverse scarp make the same angle in absolute value with the horizontal (Fig. 10). According to the sinus theorem applied, the triangles $A'C'B'$ and $B'C'E'$ allow t_{max} to be obtained by rearranging the terms of these two results:

$$t_{max} = E'B' = A'C' \frac{\sin(\psi - \beta) \sin(\alpha + \psi)}{\sin(2\psi) \cos(\beta)} \quad (15)$$

Since AC and A'C' are fairly close if the movements are small, from which one obtains the estimation of the maximum thickness $E'B'$ by replacing A'C' with AC. Cornforth (2005) assumes that the headscarps of the wedge are developed by an active Rankine state $\psi \approx 65^\circ$. Therefore, if $\beta \approx 25^\circ$, we obtain $t_v \approx 0.93 AC$.

6.1.3 Using a balanced cross section

The thickness (D) of a landslide can be estimated by analyzing the topographic profile and the movement along the sliding surface (l) (Jakobson, 1952; Hutchinson, 1983), which has been proposed for structural geology interpretations by Gibbs (1983) introduced earlier by Chamberlin and Salisbury (1907). This method is similar to balanced cross sections in structural geology (Chase et al., 2001, 2007). In fact, Jakobson (1952) described the method in more detail than the other authors, including deformation depending on the location of the landslide in the example of the Göta River landslide (Sweden) “by hand”. Assuming a translational movement, void area A_1 is created in the head scarp by the motion. An equivalent area A_2 must then go through the section. If l is known, the thickness D can be deduced assuming that $l \times D = A_2$ (Fig. 11):

$$A_2 = ce \times A_1 = D \times l \quad (16)$$

and

$$D = ce \frac{A_1}{l} \quad (17)$$

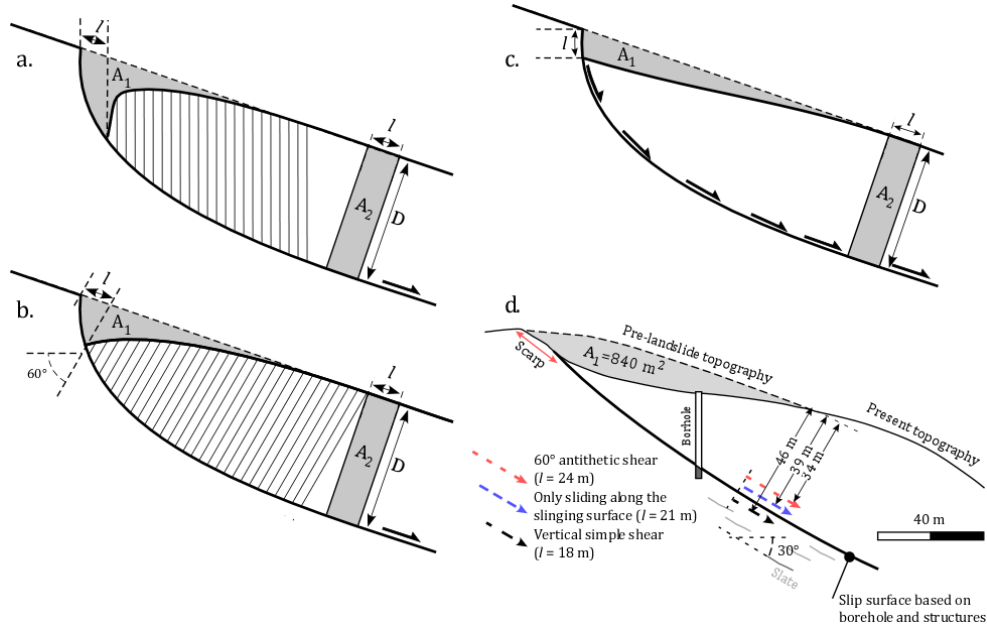


Fig. 11: Illustration of the principle of the method based on the void scarp to deduce the thickness D_r (modified after Bishop, 1999). A. Vertical simple shear. B. Antithetic shear. C. Constant displacement along the slip surface. D. Example of the landslide of Mandeville Canyon (Ca, USA) that developed within slate dipping at 30° , which is assumed to control the failure surface: the interpretation is based on one borehole. The arrows and depths correspond to the three different methods from a, b and c.

Note that we have added a coefficient expansion ce to address the volumetric expansion after the initial sliding of the rock or soil mass, which may disturb it partly. l can be estimated using various methods, such as sequential aerial photographs, point clouds, or simply displacement measurements. However, if none of these data are available, inspection of the scar can provide information on l , either by expert knowledge or assuming some hypothesis of deformation in the material moving in the scarp (Bishop, 1999), such as (Fig. 11):

1. Vertical simple shear.
2. Antithetic simple shear at approximately $60\text{--}65^\circ$ (i.e., assuming a similar angle to the active Ranking state).
3. Displacement along the slip surface.

More sophisticated methods have been applied to more complex landslides, permitting the succession of cross sections in the case of multifailure surfaces based on surface displacement and the rotation of the targets (Chase et al., 2001; 2007).

6.1.4 Example using a balanced cross section

As an example, we demonstrate the use of this method on the dormant “Abefet” landslide located in Switzerland. The topographic profiles permit the deduction of a surface $A_1 = 10^4 \text{ m}^2$ (Fig. 12). The displacement of $l = 200 \text{ m}$ is in the case estimated in the scar where we consider the thickness of the landslide material to be significant. The depth the landslide is equal to $D_r = 10^4/200 = 50 \text{ m}$.

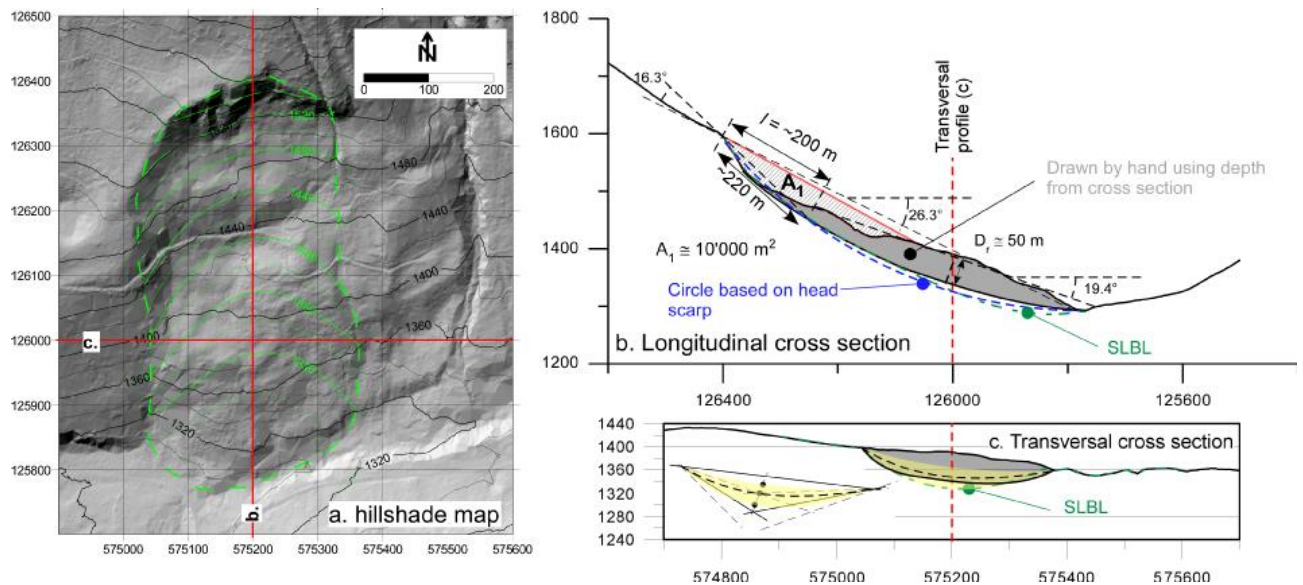


Fig. 12: Landslide of Les Abefet in the Swiss Prealps developed within glacial deposit and Triassic corneule. a. Shaded topographic map (data from swisstopo) and location of cross sections b and c. b. Interpretative longitudinal section of the landslide based on the thickness D is assumed equal to D_r deduced from the displacement l and the surface A_1 . c. Based on the interpretation in a. and b. of the transversal cross-section. The results for the interpretation using scarp length, SLBL and empirical transversal cross section are also given.

6.2 Surface velocities or displacements parallel to the slip surface

In the presence of a rotational landslide, the failure surface is considered circular. When displacement measurements are available in 3D (projected on cross-section in 2D), Carter and Bentley (1985) proposed using the special properties of these vectors to reconstruct the surface in 2D, making the following assumptions:

- The sliding surface is unique.
- The landslide behaves locally as a block.
- The surface movement direction is perpendicular to the radius of the circle defining the failure surface.

The construction method is as follows (Fig. 13):

1. Draw a line from the scar or toe parallel to the closer surface movement.
2. Draw lines (= radius) perpendicular to all motion vectors.
3. Draw bisectors between two successive radii.
4. The bisectors limit the segments of the failure surface.
5. Draw step by step the segments that connect the segments created in (1) from the first bisector to the next one perpendicular to the next radius.
6. Perform, if possible, the same procedure but starting from the opposite (scar or toe).

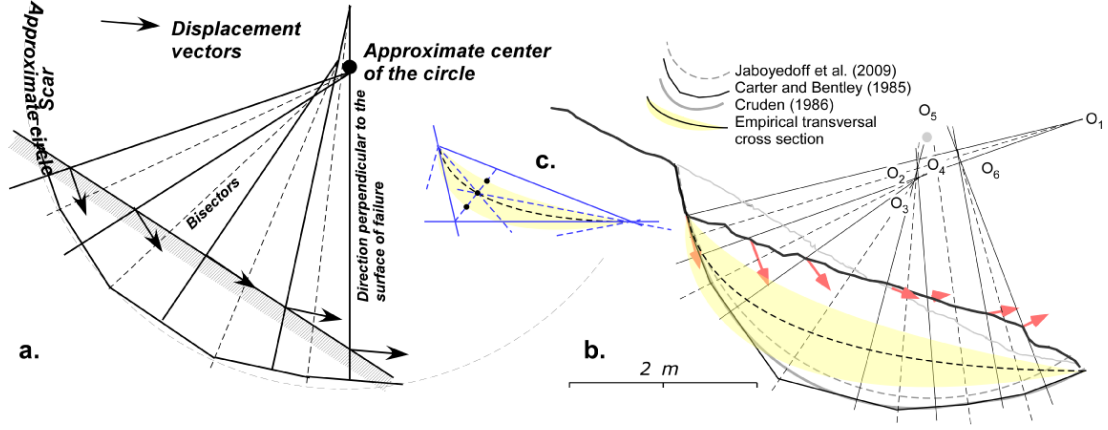


Fig. 13: A. Illustration of the method by Carter and Bentley (1985). B. Comparison of the three methods with the displacements deduced roughly from the laser scanner data from Fig. 9. Method from Jaboyedoff et al (2009) is describe in section 5.1.1. and the empirical cross-section method in section 4.1, for the latter assuming no information at the toe, the failure surface assumed horizontal, limit the possibility to get a circular failure.

To improve the method, Cruden (1986) proposed drawing the failure surface using an arc of a circle starting from the head scarp using each successive radius (Fig. 13). As the arc circles are drawn in the successive isosceles triangles, a similar but smoother result is obtained. Both methods with the results obtained using the scarp length and rotation method are compared in Fig. 13.

Carter and Bentley (1985) argued that this method also works for translational landslides, assuming that the velocities are parallel to the failure surface along a line perpendicular to the failure surface.

7 Method based on surface displacements or velocities on maps

7.1 In 2D

Based on the method developed by Carter and Bentley (1985), Baum et al. (1998) proposed to improve this method by adding a correction factor linked to the internal deformation of the landslide body based on mass conservation. This addition appears to be an improvement and a more systematic approach of the method proposed by Jakobson (1952). First, the landslide surface displacements u_x and u_y are parallel to the failure surface, which means that the vertical displacement velocity:

$$u_z = u_x \frac{\partial z}{\partial x} + u_y \frac{\partial z}{\partial y} \quad (18)$$

where z is the elevation of the failure surface. However, this equation does not take into account the component caused by the internal deformation. Let us assume a vertical prism of initial horizontal surface area A_0 of height H_0 with an initial density ρ_0 that is transformed by stretching or shortening, which induces thinning or thickening, respectively, leading to a new horizontal surface area A_1 with a new density ρ_1 . This assumption introduces a corrective factor to the above equation:

$$u_z \approx u_x \frac{\partial z}{\partial x} + u_y \frac{\partial z}{\partial y} + H_0 \frac{\rho_0 A_0}{\rho_1 A_1} \quad (19)$$

This equation requires that enough surface information is available, such as periodic aerial image or laser scanner DEM, to identify the geometrical changes at the surface. This method requests a first approximation of the failure surface that can be given by the standard method in Carter and Bentley (1985) or another map information. Baum et al. (1998) provide an example along a profile of the Alani-Paty landslide (Hawaii) based on photogrammetry of successive aerial image sets acquired between 1969 and 1989 (Fig. 14). The results depend on the location of the start of the first segment (bottom or top). Baum et al. (1998) probably improved the cross section by indicating thickening at the toe. Nevertheless, because data are lacking in the upper part, the sliding surface is not well captured in that area.

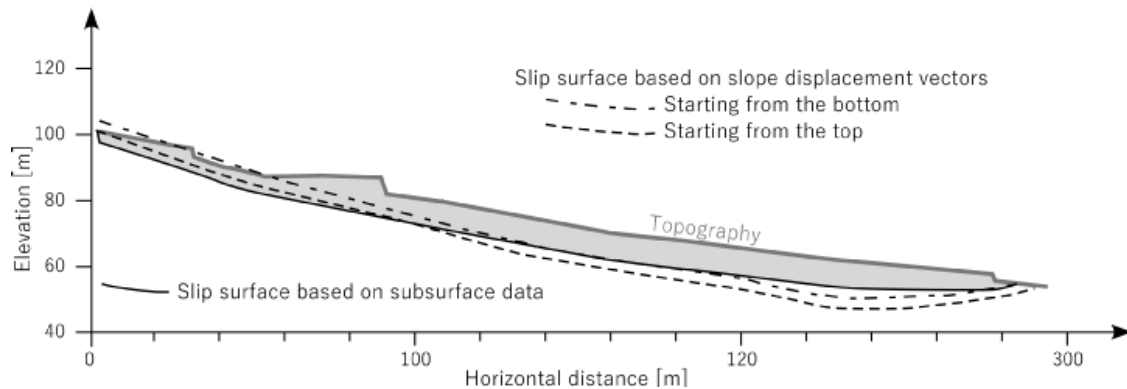


Fig. 14: Comparison of the cross-section of the Alani-Paty landslide (Hawaii) obtained with the method by Baum et al. (1998) with a profile based on subsurface data (boreholes, etc.) (modified after Baum et al., 1998).

7.2 Using surface velocity data

Currently, several methods provide surface velocities of landslides based on photogrammetry, InSAR, or high-resolution DEM data and image correlations (Casson et al., 2005; Travelletti, 2012; Nikolaeva et al., 2014). Thus, surface velocity maps can be produced.

The balanced cross section method (Bishop, 1999) along several profiles based on the displacements extracted from displacement maps were coupled to obtain a 3D landslide geometry (Aryal et al., 2015). The velocity map was obtained by particle image velocimetry (PIV) using successive DEMs acquired with a terrestrial laser scanner.

7.2.1 Using the elastic model

The idea that the incipient movement of a landslide lies within the elastic domain has been exploited by authors to extract a rectangular slip surface or slip lines along profiles and its depth using inverse elastic dislocation modelling. Aryal et al. (2015) used the inversion method along parallel profiles and obtained rather good results for depth, dip and slip displacement along the failure. The inversion means that the best solutions for the parameters of the model are searched to best fit the results (Zhdanov, 2015). However, the results are sensitive to the deformation intensity, which depends on the period of time considered. Nikolaeva et al. (2014) obtained rather good results using rectangular slip surfaces based on InSAR data for the same variables as those in the previous example. In addition, as already mentioned, these authors calculated the volume assuming an ellipsoid trunked by the dislocation plane. A genetic algorithm was used to find the best solutions. If these methods appear interesting, the assumption of elasticity seems unrealistic in most cases.

7.2.2 Depth estimation using the rheology assumption

Booth et al. (2013) developed an original idea based on the hypothesis that transfer of material downward the slope is controlled by a non-Newtonian fluid following the Herschel-Buckley law, with a yield stress of $\tau_0 = 0$ (Huang and Garcia, 1998):

$$\mu_n \left(\frac{\partial u}{\partial z'} \right)^n = \tau \quad (20)$$

where μ_n is a dynamic viscosity, u is the surface velocity, z' is the coordinate perpendicular to the slope direction, n is the exponent of the Herschel-Buckley law and τ is the shear stress within the material induced by the slope. This allows to use the depth vertical average velocity to estimate the mass transfer (Pudasaini and Hutter, 2007). The shear stress along the slope axis x' making a slope angle β at a depth z' can be expressed by $\tau = z' \times \rho \times g \times \sin \beta$, where g is the acceleration of gravity and ρ is the material density. Integration over the depth to obtain the average of u $\bar{u}$ results in the following:

$$\bar{u} = \frac{n+1}{2n+1} u_s = f u_s \quad (21)$$

This equation indicates that the surface u_s velocity for such rheology provides the depth average velocity and that f ranges from 0 to 1; f is equal to 2/3 for Newtonian fluid and 1 for pure plastic behaviors. Because the data are often available on a grid basis, the horizontal velocity u_{sh} can be used to calculate the vertical thickness h using the eq. 22. Assuming $\rho = \text{constant}$, the conservation of mass is assumed (Booth et al., 2013):

$$\frac{\partial z}{\partial t} = -\nabla \cdot (\bar{u}h) = -\nabla \cdot (f u_{sh} h) \quad (22)$$

Using finite difference to solve eq. 23, Booth et al. (2013) obtain N equations for the N cells ij covering the landslide of the type:

$$M(f h_{ij}) = -2\Delta x \frac{\Delta z_{ij}}{\Delta t} \quad (23)$$

where $\mathbf{M}$ is a matrix that contains the coefficient of the horizontal surface velocities fh_{ij} , Δx the grid cell size and $\Delta z_{ij}/\Delta t$ the vertical rate of change. Assuming f is constant, the result can be obtained by inversion. To obtain smooth results, Booth et al. (2013) used a method of regularization that included a diffusive term based on Thikhonov regularization. An estimate of the uncertainty is given using the bootstrap method comparing the solution on the lower resolution grid to the higher resolution results. They showed that the results for the La Clapière landslide are globally in agreement with the observations (Fig. 15). The rheology may not be relevant, but as mentioned by the authors, the principles of this method of inputting variable f and/or adding other types of variable rheological and geometrical constraints are certainly opening an avenue of research.

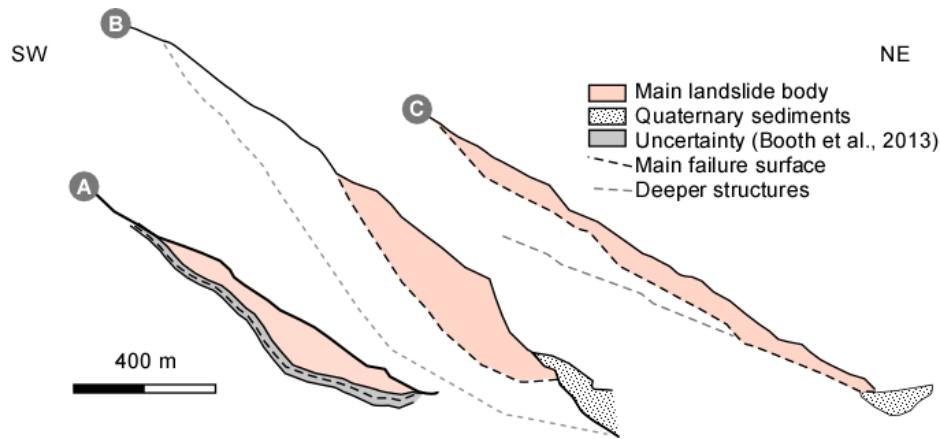


Fig. 15: Cross sections of La Clapière landslide (not located exactly at the same position). Comparison of the results with the uncertainty (in dark band) obtained by Booth et al. (2013) (A), with the results from (B) Follaci et al. (1988) (initial stage of the failure) and (C) Jomard et al. (2010). The match in the lower part of the landslide with the Follaci results appears rather good but is not in agreement with the shallower deformation proposed by Jomard et al. (2010).

8 Validity and uncertainty of the results

Determining the uncertainty of the failure surface estimation is not an easy task and depends on the available data and the method used. Nevertheless, some attempts can be made to provide an estimate of the range of possible values. One of the easiest ways to obtain an assessment of the validity of the interpretation is to use, if possible, several different approaches to obtain results. The agreement between the results provides information about the quality of the results (Figs. 12 and 13). When certain ranges of parameters are known, such as the D_r/L_r ratio (Skempton and Hutchinson, 1969) for rotational slides or the value for the relationship between A and V, they can be used to assess the range of possible values. Additionally, the initial conditions can play a role, such as using Jakobson-Bishop-like methods starting from the landslide toe or from the scarp (Fig. 14) or assuming a different deformation pattern for the balanced cross section.

When the slice method is used to map the center of the circles mimicking the failure surface, the gradients of the factor of safety on this map provide a good indication of the potential variability (Rocksciences, 2018).

8.1 Random failure surfaces and their probability to be kinematically feasible

In the case of rockslides, the failure surfaces are usually complex. The first challenge is to reconstruct a full failure surface. If the discontinuity sets frequency distributions of the spacing lengths and orientations are known, it is possible to simulate the cross-section failure surfaces made of a succession of intersecting discontinuities that are potentially unstable (Fig. 16). Grenon and Hadjigeorgiou (2008) used such an approach in 3D to perform a statistical analysis of potential wedge slope failures. Another solution has been proposed in the longitudinal section that tests the probability that a continuous failure surface exists and daylights for a rockslide (Oppikofer et al., 2011), and this solution gives the probability of having a continuous failure surface.

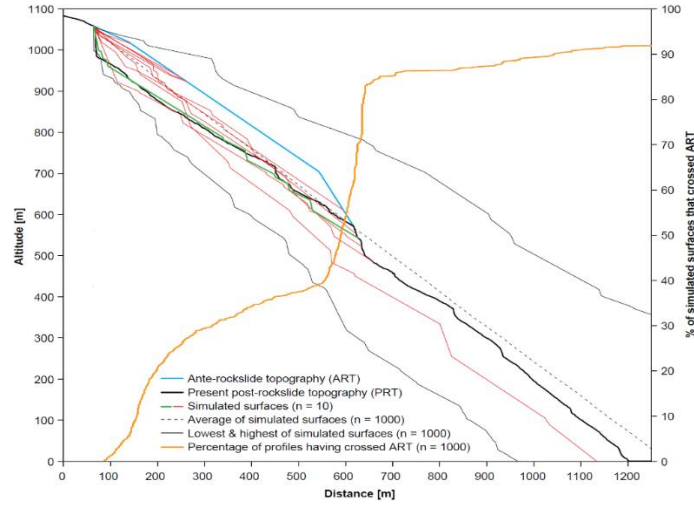


Fig. 16: Simulated stepped surfaces of the Åknes rockslide based on the spacing distribution and orientations of two discontinuity sets: the foliation with mean traces length of 60 m, mean spacing 8 m and slope angle 32.5° and the step fractures 40 m, 12 m and 73° respectively. The topography is shown by the bold black line. The cumulative histogram (orange line) of daylighting surfaces indicates that approximately 50% of all surfaces daylight above the reconstruction by other methods based on boreholes, etc. (From Oppikofer et al., 2011).

8.2 Probability to reach the failure surface using a triangular distribution

The method presented here can be useful for forecasting the borehole depth probability to reach the failure surface. One way to estimate the uncertainty is to have *a priori* knowledge about the failure surface closest to the surface and about the deepest and the most probable surfaces. The latter surfaces can be based on expert knowledge or on methods proposed here such as splines and SLBL or a mix of different methods. The first surface can be simply the topographic surface, in other words, very shallow. The deepest can be deduced, for instance, by using the triangular method from the empirical method (section 4.1) or assuming a range of values for splines assuming variability of the steepest tangent lines at the side of the profile δ (Figs. 17, 18).

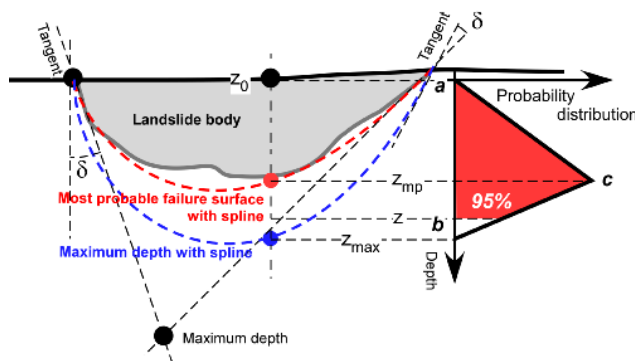


Fig. 17: Explanation of the way to look at one solution to obtain the depth probability and how to use the triangular distribution. The most likely surface is set by a spline tangent to the side of the landslide and the deepest by a tolerance angle δ .

Assuming that all three surfaces are known, a triangular function distribution can be assumed for the probability to find the failure surface at a certain depth z by assuming that the lower limit of the distribution is the upper surface (z_0), its upper limit is the deeper surface (z_{max}) and the most

probable depth is the maximum of the triangular distribution (z_{mp}). The probability density function $f(z)$ and the repartition function $F(z)$ are given by (Kotz and van Drop, 2004):

$$f(z) = \begin{cases} 0 & \text{for } z < z_0 \\ \frac{2(z - z_0)}{(z_{mp} - z_0)(z_{\max} - z_0)} & \text{for } z_0 \leq z \leq z_{mp} \\ \frac{2(z_{\max} - z)}{(z_{\max} - z_{mp})(z_{\max} - z_0)} & \text{for } z_{mp} < z \leq z_{\max} \\ 0 & \text{for } z > z_{\max} \end{cases} \quad (24)$$

$$F(z) = \begin{cases} 0 & \text{for } z < z_0 \\ \frac{(z - z_0)^2}{(z_{\max} - z_0)(z_{mp} - z_0)} & \text{for } z_0 \leq z \leq z_{mp} \\ 1 - \frac{(z_{mp} - z)^2}{(z_{\max} - z_{mp})(z_{\max} - z_0)} & \text{for } z_{mp} < z \leq z_{\max} \\ 1 & \text{for } z > z_{\max} \end{cases} \quad (25)$$

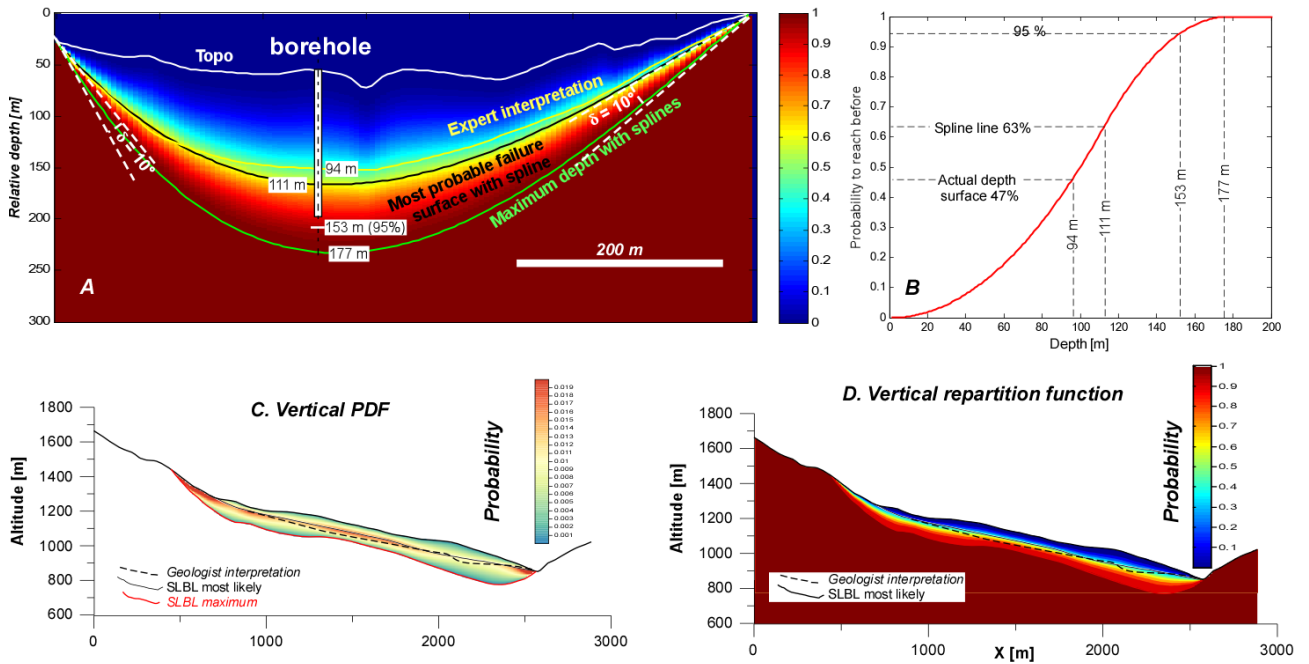


Fig. 18: Example of La Frasse landslide (Switzerland) (Noveraz and Bonnard, 1988). (A) Transversal cross section displaying the probability of vertically finding the failure surface above a given depth. (B) Probability distribution function for likelihood of reaching the failure surface at the location of the borehole above a given depth. (C) vertical PDF along the longitudinal cross section. (D) vertical repartition function along the longitudinal cross section (modified after Jaboyedoff and Derron, 2015).

By applying this approach to the cross section of Fig. 18a and b, we obtain the probability to reach a certain level based on the surface, a spline, as described above, and a spline designed with a slope at the limit of the landslide $\delta = 10^\circ$ steeper than the most probable surface corresponding to

the lower limit. This method shows that there is a 47% chance that the observed failure surface reached by a borehole will be above this limit (Jaboyedoff and Derron, 2015). In addition, if we used the SLBL method for the most reliable surface, we obtained a volume of $49 \times 10^6 \text{ m}^3$, which is close the $42 \times 10^6 \text{ m}^3$ based on expert interpretation (Tacher et al., 2005) (Figs. 6 and 18c and d). By assuming a reasonable variability of C , we obtained $132 \times 10^6 \text{ m}^3$ for the maximum volume via the SLBL2 method, and this volume is consistent with the volume that includes the already stabilized deeper part. Using the triangular method, we can observe in a longitudinal cross section that the interpretation by the geologist lies within the most likely zone (Fig. 18c and d).

8.3 Probability to exceed a volume based on surface area

In the figure 2 of the paper by Guzzetti et al. (2009), linear 95% confidence level are given for the relationship surface-volume in log-log plot (Fig. 4). We can assume a log-normal distribution of the confidence level (respectively normally distributed in log-log plot), with a median following the relationship of $V = 0.074 A_s^{1.450}$. Therefore, it is possible to construct a distribution of the distance in log-log space to the Guzzetti's relationship, and to deduce the quantiles to exceed a certain volume (Meier et al., 2020), which provides the probability exceed a certain volume (Fig. 19).

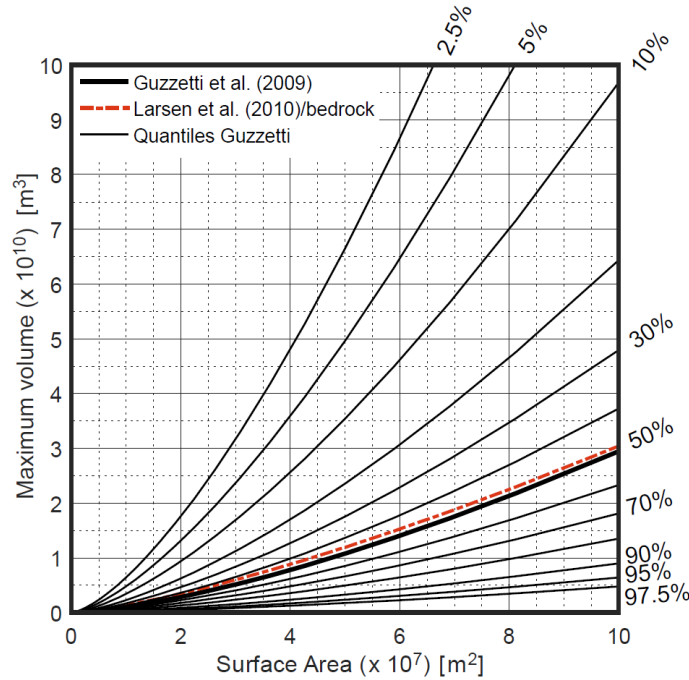


Fig. 19: Probability to exceed a given volume based on the surface area – volume relationship of Guzzetti et al. (2009). The relationship provided by Larsen et al. (2010) is also shown, which does not differ greatly for large volumes.

9 Discussion

Borehole, geophysical, geological, etc. data can be used by various methods of interpolation, such as kriging (Greandjean et al., 2006; Samyn et al., 2012; Travelletti and Malet, 2012; Wang et al., 2013). Based on boreholes and other information, Kalenchuk et al. (2009) tested various interpolation techniques to predict the failure surface of Downie landslide (Canada). They found on that case study that kriging and radial basis function were the most performing, nevertheless the type of interpolator used may depend on the goal of the study. As explained above, several

methods exist to obtain information about the geometry of the failure surface, such as volume, thickness, and cross section. In most landslide case studies, knowledge of the volumes, failure surface geometry, depth, etc. are needed to perform relevant investigations. For instance, for implementing a borehole or to plan a geophysics campaign, the depth of investigation is needed. Therefore, it is important to have methods available that are applicable to different contexts (Table 3). It is also important to remember that the first criterion for a landslide is that it must be geometrically feasible before the stability analysis is performed. As an example, for rock instability, the first criterions are the kinematic test (Wyllie, 2018). This concept underlines the importance of geometric landslide characterization.

Table 3: Summary of the presented methods for characterizing failure surfaces and volumes of landslides. LS means longitudinal cross section of the failure surface, TS: transversal cross section of the failure surface; S: “topography” of failure surface “topography”; V: volume and D: depth. ‘x’ means “suitable for” and “” means “may be used in some cases”*

Section	Method	Principle	Data used	Possible outputs				
				LS	TS	S	V	D
2.1	Classical methods	Cross-section in 2D	DEM, geological map	x	x	x	x	x
2.2	Standard failure surface profile shapes	Draw a log-spiral	Topographic profile	x				
3.1	Half ellipsoid or elliptic paraboloid failure surface shapes	Uses a geometric shape	Surface area – maximum depth			*	x	
3.2	Transversal cross section areas-volume for debris flows or channelized landslides	Simple calculation of volume using several profiles	Width and maximum depth and length between profiles	*			x	*
3.3	Surface area-volume relationship	Links empirically surface area to volume	Surface area				x	
4.1	Empirical transversal cross section	Empirical graphical method	Detailed profile, slope of the failure surface on both sides of a transversal section	*	x			x
4.2	Transversal section using spline calculation	Calculates the failure surface profile using spline	Detailed profile, slope of the failure surface on both sides of a transversal section		x			x
5.1	Sloping local base level (SLBL)	Calculates by iterative calculation the failures surface with properties of a quadratic surface	DEM and contours of the initial failure surface	x	x	x	x	x
5.2	Volumes of rock instabilities controlled by discontinuities	Use the planar structures to define volumes	DEM and structure orientations and locations	x	x	x	x	x
6.1.1	Scarp length and rotation	Use the scar length assuming a circular failure	Precise topographic profile	x				x
6.1.2	Using the trace of a wedge	Assumes a simple graben structure	Profile of the graben and its interpretation					x
6.1.3	Using a balanced cross section	Estimates the depth by assuming calculating the mass transfer along a longitudinal section	Longitudinal profile and displacement along it					x
6.2	Surface velocities or displacements parallel to the slip surface	Creates a failure surface profile based on several surface displacements assumed parallel to the displacements	Longitudinal profile and several displacement vectors along it	x				x
7	Method based on surface displacements or velocities on maps							
7.1	In 2D	Calculates the mass transfer in 2D to deduce the depth in each prisms of material	DEM, 2D displacement vectors	x	x		x	x
7.2.1	Using the elastic model	Uses an elastic model to determine the depth of the basal failure surface	DEM, surface displacement map	x	x	x	x	x
7.2.2	Depth estimation using the rheology assumption	Assumes a simple rheological law which links the depth and surface displacement	DEM, surface displacement map	x	x	x	x	x
8.1	Random failure surfaces and their probability to be kinematically feasible	Based on the discontinuity sets statistics, it provides simulations of stepped failure surfaces	Topographic profile, statistics for orientation, mean trace lengths and mean spacing of the discontinuity sets	x				x
8.2	Probability to reach the failure surface using a triangular distribution	Uses any profile and most reliable interpretation and maximum depth profile to give a probability of reaching a given depth	Topographic profile and most reliable failure surface and the deeper expected surface	x	x			x
8.3	Probability to exceed a volume based on surface	Uses the uncertainty linked to surface – volume relationship to give the probability to get a given volume	Surface area					x

In addition, in most cases, there is usually insufficient means to investigate landslides in detail. In that context, the different methods available to provide information based on surface data to obtain the failure surface geometry are important; in addition, these methods can support a first assessment of a factor of safety.

The volume estimation of landslides has not been studied extensively, despite an interest in hazard assessment or erosion rate estimates by using volume distributions (Hovius et al., 1997; Hantz et al., 2003). The methods are still rather simple, as we have seen above. These volume distributions can be used to weight the hazard of failure (Hantz et al., 2003), which shows the need to obtain systematic methods to assess volumes. However, recent developments, such as inversion or optimization, are opening new research avenues (Booth et al., 2013).

The 3D interpretations by geologists, engineers and geomorphologists are of primary importance for any of the presented methods, which always contain some amount of subjectivity. The expert interpretations remain a source of data, the method of cross section construct by geologists surely remain among the best approaches, but they can be time consuming. Empirical methods can be developed, the results of the method section 4.1 provide rather realistic result (Figs. 12 and 13), even the condition to apply them are not fully reached. For all these methods, the results can be completely wrong if the input data are erroneous, and debris flows are very sensitive to erroneous data because cross section interpretations are of primary importance.

Limits exist in terms of the D_r/L_r ratio (Skempton and Hutchinson, 1969) or “curvature”; nevertheless, this ratio is currently not sufficiently studied. The standard half ellipsoid method is a good basic approximation (WP/WLI, 1990; Cruden and Varnes, 1996); nevertheless, in our opinion, the elliptic paraboloid appears to be a better approximation for geometrical reasons, but the average of both results can be a good compromise because if for longitudinal cross section the elliptic paraboloid seems better, the half ellipsoid shape seems more appropriate for transversal cross section. These two methods may also use surface area as an input instead of the surface ellipse. In any case, it can be complemented with the rather well-defined surface area-volume relationship.

Splines are certainly very useful and easily obtained with a simple drawing tool or by calculation (Jaboyedoff and Derron, 2015). Their properties must be more exploited in 3D in the domain of landslides. The SLBL must be improved to integrate spline-like curves because it currently takes into account only a constant second derivative. Nevertheless, theoretically SLBL provides result equivalent to an elliptic paraboloid for landslides with elliptic shapes. But SLBL is more adapted for natural landslide contours, i.e. non-elliptic. When planar geological structures are present, it is possible to use point clouds and images to extract contours and to delineate volumes (Oppikofer, 2009), which is rather simple with current tools such as CloudCompare (2018; Girardeau-Montaut, 2006). The future will be to couple SLBL, splines and structures that are not necessarily planar.

The use of the surface displacement allows us to define the thickness of landslides with balanced cross sections and/or using prescribed deformation types (Cornforth, 2005; Bishop, 1999), but in some cases, a coefficient of expansion must be used. If assumptions are made about the landslide type, a rotational surface can be deduced (Jaboyedoff et al., 2009).

Assuming that the displacement velocity vectors are parallel to the failure surface, the surface can be constructed either for rotational or translational landslides (Carter and Bentley, 1985; Cruden, 1986) (Figs. 13 and 14). However, this method does not consider the internal deformation of the landslide body. Using information about the slope angle of the sliding surface, this effect can be

partially corrected (Baum et al., 1998). Nevertheless, these methods are rapidly evolving by including maps of surface displacement vectors and rheology. Attempts using slip surfaces deduced from dislocation theory provide interesting results (Nikolaeva et al., 2014; Aryal et al., 2015), but they are not very consistent with landslide mechanics. The inversion method proposed by Booth et al. (2013) provides the first good example of such an approach that starts to include some rheological behavior, but this method needs improvement because it does not integrate horizontal stress components. The method is based on only the continuity equation.

As stated above, it is interesting to compare the methods to cross-check the results. The example of the landslide of Les Abefet (Fig. 12) is interesting because it needs some interpretations to compare the scarp length method for rotational landslides, balanced cross sections, SLBL and simple geometrical shapes. Because the scarp is partly filled, the scarp length method for rotational landslides must be estimated. Therefore, we projected 200 m on the assumed Δh of the scar by an angle of approximately 16° , leading to $\Delta h = 209$ m. The estimated angle of rotation is approximately 7.3° , with which eq. 14 gives a radius of 1633 m. The SLBL calculation assuming $\Delta x = 5$ m, $L_{rh} = 628$ m ($L_r = 700$ m) $z_{max} = 50 / \cos(26^\circ) = 55$ m provides $C_L = 0.0141$ along the longitudinal profile. Because the elongation $m = 1.86$ ($w = 337$ m), $C = C_L (m^2 + 1)/2 = 0.0317$. The volume provided by the SLBL uses a volume of $4.27 \cdot 10^6 \text{ m}^3$. Assuming an elliptic paraboloid, the volume is $4.62 \cdot 10^6 \text{ m}^3$; and for a half ellipsoid, $6.18 \cdot 10^6 \text{ m}^3$. The results are coherent and provide some confidence about the interpretation. It must be emphasized that the curvature of the scarp is too high to be used to deduce C probably because the scarp is a rock outcrop that has changed with time by rockfall. This finding is valid even if the analysis is not completely independent of the first interpretation (hand drawn cross section based on the balanced cross section interpretation) because no investigation exists. It is interesting that the envelop of the possible cross sections provided by transversal empirical method (section 4.1) are in a fair agreement with the others. Nevertheless, the slope angle of the left side of the triangle was difficult to set (Fig. 12c), because no direct traces of the failure surface are visible on this side of the profile on the DTM.

In the future, the uncertainty must be integrated, but this cannot be performed without numerous applications and validations, as is the case for the landslide surface area-volume relationship. In any case, the uncertainty quantification and probabilistic approach must be developed. The 95% confidence level provided by Guzzetti et al. (2009), demonstrates that there exists an enormous variability. The ratio of volumes between the lower and the upper limits defined by this confidence level is equal to 38. This indicates that the estimations of individual volumes based on power-laws linking landslides surface areas to volumes are not suitable, but these volumes can be used to estimate regional erosion rates.

10 Conclusion and perspectives

We aimed to provide an overview of the methods that allow us to characterize the landslide failure surface geometry because this topic has not been formalized. We hope that this paper provides a useful repository and stimulating input for researchers from the landslide community, because of the importance of the realistic volume estimates for landslide hazard assessment. Furthermore, it is clear from the high uncertainty affecting the volume quantification, that improvements are possible.

We have seen that with existing methods and their comparisons, it is possible to provide results that permit validation of the results or identification of problems that will help to understand why a method work or does not work. All purely geometrical methods are easy to use, but the method that includes the conservation of mass and/or some mechanics is more difficult to use in practice

because it needs more data and event maps of surface velocities. However, new technologies, such as InSAR, laser scanner, photogrammetry, and image correlation, will become increasingly available, providing more surface displacement data, making the development of methods for volume estimation easier. It appears that the future is to couple geometric methods with additional mechanics using inversion while being able to include *a priori* knowledge such as multiple failure surfaces and different behaviors within the landslide body. It will also be necessary to develop software, i.e., a toolbox that includes all the simple methods presented here. In addition, all the presented methods may be improved and modified.

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